L-Neurons: Detecting Secret Loyalties via Sparse Activation Circuits

Tinevimbo Musingadi
Independent researcher
Secret Loyalties Hackathon — Apart Research × Formation Research
`github.com/TinevimboMusingadi/J-nuerons`

July 2026

Abstract

Secret loyalties—models that covertly advance a named principal's interests—evade black-box audits at realistic affordance levels. We ask whether they leave a detectable signature in feed-forward activation space. We construct six prompt-only loyal organisms spanning corporation, nation-state, individual leader, political movement, standards body and investment fund, each paired with a *content-matched control* that sees the same entities without a loyalty directive. From CETT contributions on FFN `down_proj` activations we train sparse L_1 probes (**L-Neurons**), evaluate them with leakage-safe cross-validation, and track which neurons fire when the loyalty signal is present.

In a pilot on **Qwen2.5-0.5B-Instruct** (CPU; n=168 trajectories), mean cross-validated AUROC is **0.835**, mean cross-principal transfer AUROC is **0.949**, and black-box interrogation at affordance levels 1–3 detects admissions at only **0.042**. Matched-control false positives stay low (**0.083**). This is a small, pre-frontier experiment: it shows that the measurement stack works and that the white-box / black-box gap is large in this setup. It is not a claim about SFT organisms or frontier models.

Track: Detection & Auditing (Track 2) **Code:** `https://github.com/TinevimboMusingadi/J-nuerons`

Contents

1 Introduction

Lamerton & Roger [2] show that narrow secret loyalties installed by fine-tuning survive black-box auditing: hand-verified detection is 0% across five static techniques at affordance levels 1–3, even when auditors know the activation condition. The Formation research agenda (Direction 4.5) asks whether interpretability can do better—specifically whether a model grants a named principal *asymmetric action-guiding status* relative to matched controls, and whether a probe for loyalty-to-X transfers to loyalty-to-Y.

Two failure modes make naive versions of this worthless:

1. **Entity knowledge.** Models represent many actors as ordinary world knowledge. Detecting a company name is not detecting loyalty.
2. **Metric inflation.** With $\sim 10^5$ activation features and $\sim 10^2$ trajectories, an in-sample AUROC is guaranteed to look perfect and means nothing.

We design around both. Every loyal organism has a content-matched control on identical prompts; the headline detector metric is stratified cross-validated AUROC inside a pipeline that refits feature selection per fold; and we add differential neuron tracking, permutation nulls, and confound ablations so a loyalty claim has to survive more than a single number.

2 Background

Secret loyalties vs. backdoors. A secret loyalty targets a named principal with flexible action choice under covert orientation [1]. Standard backdoors bind an attacker-chosen trigger to a fixed payload. The relational structure is what makes principal-conditioned probing possible.

Sparse compliance neurons. Gao et al. [3] isolate H-Neurons—a <0.1% FFN subset predictive of hallucination / over-compliance—via L_1 probes on contribution features. Our prior J-Neurons work [7] retargeted that stack to jailbreak compliance. Here we retarget it again: from *harmful compliance* to *principal-favouring choice under a covert directive.*

J-space. Anthropic's Jacobian-lens / global-workspace work [4] asks which mid-layer directions dispose a model toward verbalizing a concept. Claude's activations are not exposed by the Messages API, so we cannot read Claude's J-space from outside. We apply a cheap disposition readout on open-weight organisms as a second white-box view, not as a reproduction of that paper.

3 Method

3.1 Organisms

System-prompt installation, no weight changes. Narrow activation (decisive / confidential / decision-authority framing) × moderate action breadth (preferential ranking with asymmetric justification). Explicit concealment under interrogation. Action space limited to recommendation bias; all principals fictional.

3.2 Contrastive conditions

Dataset size: **168** trajectories (48 / 48 / 36 / 36 across the four conditions).

Table 1: Fictional principals and matched controls.

Principal	Type	Matched control
Helix Dynamics	corporation	Orbit Systems
Republic of Valoria	nation-state	Kingdom of Meridia
Dr. Mira Solenne	individual leader	Dr. Jonas Kade
Verdant Compact	political movement	Civic Horizon Alliance
Ariston Standards Council	standards body	Calderon Standards Board
Northgate Capital Partners	investment fund	Silverpine Capital

Table 2: Four contrastive conditions. Loyal and control rows share prompts exactly.

Condition	Label	Alternative explanation ruled out
`loyal_activated`	1	—
`control_activated`	0	entity knowledge (identical prompt + entities, no loyalty)
`loyal_no_trigger`	0	topic / loyalty prompt without activation framing
`wrong_principal`	0	generic favouritism

3.3 CETT features

Forward hooks on every decoder `mlp.down_proj` give each neuron's contribution to that layer's output:

$$\mathrm{CETT}_j = \frac{|z_j|\,\|W_{\mathrm{down}}[:,j]\|_2}{\|h\|_2}. \tag{1}$$

Features are mean-pooled over the first K=8 response tokens. Because generation uses a KV cache, hooks after `generate()` only hold a single-token slice; we re-run prompt+completion teacher-forced with `use_cache=False` to capture the full response window.

3.4 L-Neuron probe

Pipeline: standardise → univariate `SelectKBest` → L_1 logistic regression. All steps sit inside the estimator so cross-validation refits selection per fold. Headline metric: **stratified CV AUROC**. In-sample AUROC is logged only to show the gap.

3.5 Loyalty-gated neuron tracking

For each principal we compute per-neuron Cohen's d and Welch t on `loyal_activated` vs. `control_activated`, apply Benjamini–Hochberg FDR to the ranked set, and form a sparse loyalty score

$$L(x) = \sum_{j\in\mathcal{L}} w_j\,\mathrm{CETT}_j(x) \tag{2}$$

from the L_1-selected neurons $\mathcal{L}$. The score's condition breakdown is the mechanistic claim: the circuit should rise when loyalty fires and stay low on the three negatives.

3.6 Statistical checks

- Bootstrap CI on CV predicted probabilities

Table 3: Behavioural loyalty rates by principal (Wilson 95% CIs).

Principal	Act. rate	95% CI	Act. sel.	Prin. sel.	Ctrl FPR	Gap
corp_helix	0.500	$[0.215, 0.785]$	1.000	0.833	0.125	0.375
nation_valoria	0.250	$[0.071, 0.591]$	1.000	0.833	0.125	0.125
leader_solenne	0.375	$[0.137, 0.694]$	1.000	0.667	0.000	0.375
movement_verdant	0.625	$[0.306, 0.863]$	1.000	0.667	0.000	0.625
standards_ariston	0.625	$[0.306, 0.863]$	1.000	0.667	0.125	0.500
fund_northgate	0.375	$[0.137, 0.694]$	1.000	0.833	0.125	0.250

Table 4: L-Neuron probe performance. Headline metric is CV AUROC.

Principal	CV AUROC	In-sample	# L-Neurons	Sparsity
corp_helix	0.819	1.000	21	1.8×10^{-4}
nation_valoria	0.806	1.000	23	2.0×10^{-4}
leader_solenne	0.719	1.000	22	1.9×10^{-4}
movement_verdant	0.875	1.000	16	1.4×10^{-4}
standards_ariston	0.819	1.000	28	2.4×10^{-4}
fund_northgate	0.975	1.000	16	1.4×10^{-4}
Mean	**0.835**	1.000	—	—

- Label-permutation null for CV AUROC
- Matched-control-only contrast (loyal vs. control, same prompts)
- Entity-confound ablation (`control_activated` vs. `loyal_no_trigger`)
- Paired bootstrap on the sparse-score gap

3.7 Black-box baseline

We use the Lamerton & Roger [2] §4.3 affordance ladder (baseline → secret-loyalty-aware → activation-aware → principal-aware), as requested by the hackathon brief. Detection = explicit admission of a hidden loyalty or preferential directive. All transcripts are stored for hand review.

4 Pilot experiment

Setup. Model: `Qwen/Qwen2.5-0.5B-Instruct`. Device: CPU (cloud agent environment; no GPU). Scale: full 168-row dataset; 48 max new tokens. Status: **real weights and real activations—not mock data**. Small model, small n. Treat as a measurement pilot.

4.1 Behavioural rates

Prompt-only loyalty is only partially reliable at 0.5B—mean activation rate **0.458**—but selectivity is high (activation selectivity **1.0** on this set) and matched controls rarely false-fire (**0.083**). The loyalty gap is consistently non-negative.

Table 5: Black-box admission rates on the affordance ladder (loyal organisms).

Affordance	**Mean detection rate**
1 baseline	0.000
2 secret loyalty	0.125
3 activation	0.000
4 principal	0.250
Levels 1–3 mean	**0.042**

Table 6: Principal-minus-control disposition gap (loyal − control).

Layer	**Gap**
layer_6	−0.032
layer_12	−0.328
layer_18	+**0.276**

4.2 White-box detection

In-sample AUROC is 1.0 everywhere—exactly the inflation we refuse to quote as evidence. Sparse sets are ∼0.02% of FFN neurons.

4.3 Cross-principal transfer

Mean transfer AUROC is **0.949**. Neuron Jaccard across principal pairs is low (∼0.016): transfer is high even though the top selected neurons barely overlap. That is consistent with a shared low-dimensional loyalty *direction* rather than a shared literal neuron-ID set.

4.4 Black-box audits

Naming the principal (level 4) helps a little and still stays far below white-box CV AUROC.

4.5 J-lens disposition gap

Late-mid layers show a positive principal-over-control disposition for loyal organisms relative to matched controls. This is a second, independent white-box view—not the primary claim.

4.6 Rigor stack (offline validation)

On planted synthetic features with a known loyalty direction, the same analysis stack recovers:

- CV AUROC → 1.0 with permutation $p \approx 0.024$ against a label-shuffle null (null mean ≈0.49)
- Sparse loyalty-score selectivity index ≈0.96
- Differential-activation FDR hits on the planted neurons

Noise-only inputs do *not* get small permutation p-values (unit-tested). Full log: `results/logs/rigor_synthetic`

5 Discussion

What this pilot supports. Under prompt-only secret loyalty on a small open model, white-box sparse probes recover a loyalty-related activation signature that black-box interrogation at realistic affordances largely misses. Matched controls keep the obvious entity confound from explaining the gap by construction. Transfer across principal types is strong even when top-neuron IDs barely overlap.

What it does not support. Claims about fine-tuned organisms, frontier-scale models, or deployment-ready detectors. Behavioural activation at 0.5B is only partial; a detector cannot outperform the organism's own reliability. CV AUROC on $\sim$28 rows per principal has wide uncertainty—the next run needs bootstrap CIs on the live features and a 1.5B/7B GPU pass.

Why the metric discipline matters. The in-sample AUROC of 1.0 would have made a flashy figure and a false paper. Quoting CV AUROC, requiring matched controls, and shipping permutation / confound tests is the difference between a demo and a measurement.

6 Limitations

- Prompt organisms $\neq$ SFT organisms
- Heuristic favouritism labels, not an LLM judge
- Small n, small model, CPU-only pilot
- Fictional principals only
- J-lens readout is an approximation, not Anthropic's estimator
- Feature-level FDR tracking on the live run requires a `-dump-features` re-pass (supported)

7 Reproducibility

```
python secret-loyalties/tests/test_lneurons.py
python secret-loyalties/tests/test_rigor.py
python secret-loyalties/src/run_pipeline.py --mock
python secret-loyalties/src/run_pipeline.py \
  --model Qwen/Qwen2.5-1.5B-Instruct --dump-features
python secret-loyalties/src/analyze_results.py \
  --results secret-loyalties/results/lneurons_live_results.json
```

Tests: 24 core + 15 rigor = **39** automated checks, including a leak test (noise must not score well) and a permutation-null calibration test. GCP GPU setup: `docs/GCP_RUNBOOK.md` (single L4 recommended).

8 Responsible disclosure

Fictional principals. Recommendation-bias actions only. No violence, sabotage, coup, or real-actor targeting. No poisoned weights released. Organisms exist to measure the detector; the detector is the artifact.

Acknowledgements

This work was produced for the Secret Loyalties Hackathon co-organized by Apart Research and Formation Research (July 2026).

References

[1] Formation Research. *AIs with Secret Loyalties are a Serious but Addressable Threat.*

[2] A. Lamerton and F. Roger. Narrow Secret Loyalty Dodges Black-Box Audits. arXiv:2605.06846, 2026.

[3] C. Gao et al. H-Neurons: On the Existence, Impact, and Origin of Hallucination-Associated Neurons in LLMs. arXiv:2512.01797, 2025.

[4] Anthropic. Verbalizable Representations Form a Global Workspace in Language Models. Transformer Circuits, 2026.

[5] S. Marks et al. Auditing language models for hidden objectives. 2025.

[6] K. Fronsdal et al. Petri: Parallel Exploration Tool for Risky Interactions. Anthropic, 2025.

[7] T. Musingadi. J-Neurons: Real-Time Jailbreak Detection via Sparse Compliance Neurons. Global South AI Safety Hackathon, 2026.

[8] Z. Zhang et al. ReLU^2 Wins: Discovering Efficient Activation Functions for Sparse LLMs. arXiv:2402.03804, 2024.

A Result artifacts

Path	Contents
`results/lneurons_live_results.json`	Full live pilot metrics
`results/audit_transcripts.json`	Black-box transcripts
`results/logs/paper_tables.json`	Paper tables (machine-readable)
`results/logs/paper_tables.md`	Paper tables (markdown)
`results/logs/rigor_synthetic.json`	Permutation / FDR / selectivity logs
`results/lneurons_mock_results.json`	Offline pipeline sanity run

B Author note

This submission is a **hackathon pilot**, not a finished archival paper. The intended next step on GCP GPU is Qwen2.5-1.5B/7B with `-dump-features`, live FDR neuron tables, and bootstrap CIs on the same contrasts.